



EXAM PAPERS PRACTICE

Boost your performance and confidence
with these topic-based exam questions

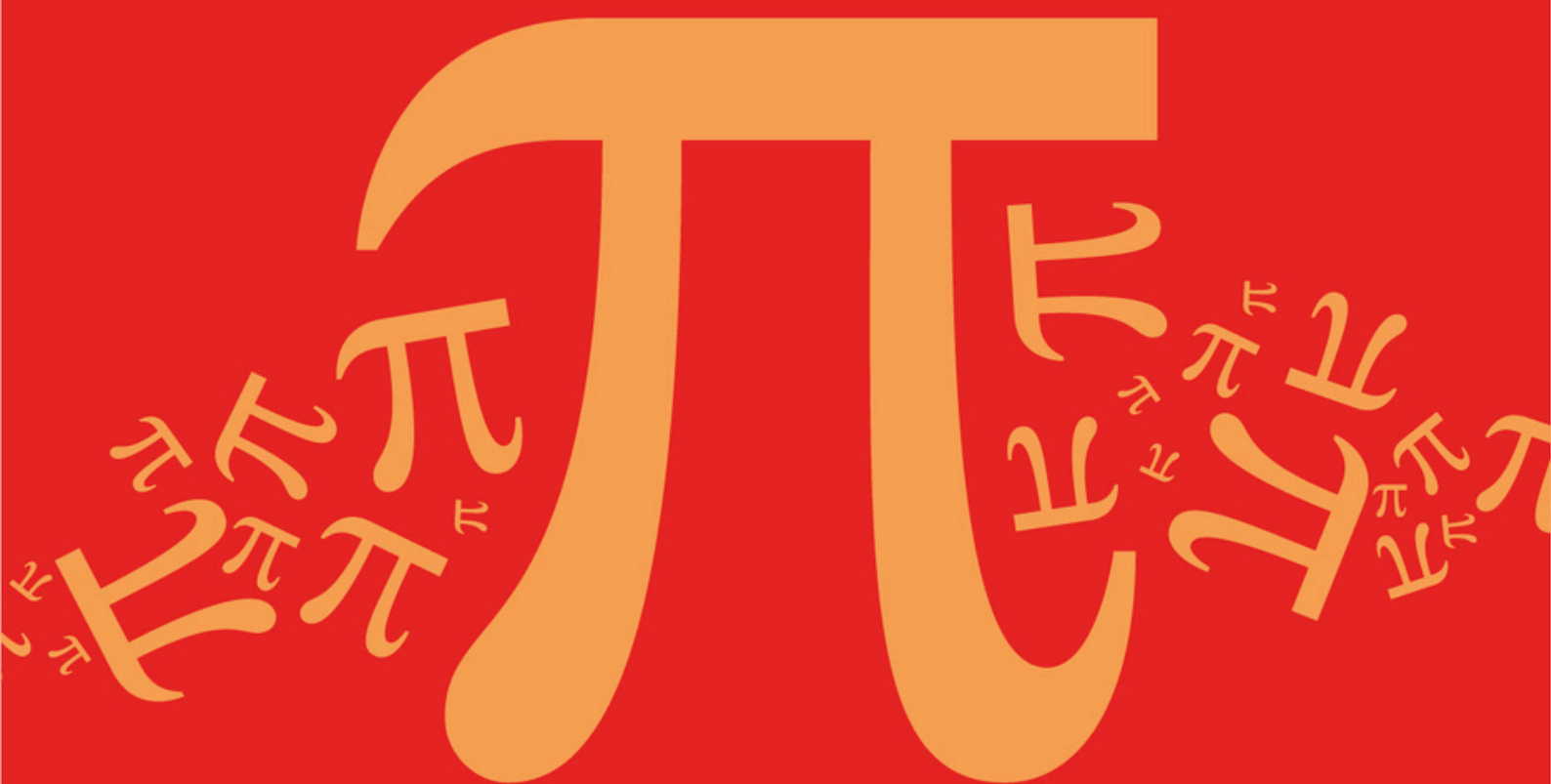
Practice questions created by actual
examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and
thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

DA: Graphs

Sub topic

DA7: Graph Isomorphisms

Mark Scheme

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Mark Scheme

Topic

DA: Graphs

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DA7: Graph Isomorphisms

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

- (a) E and H

Deduct 1 for error or omission

B2,1,0

2

- (b) F, G and H

Deduct 1 for error or omission

B2,1,0

2

- (c) E, F and G

Deduct 1 for error or omission

B2,1,0

2

- (d) F and G
(or M1)

A1)

2 cao, or sensible working towards wrong answer will earn ½

B2

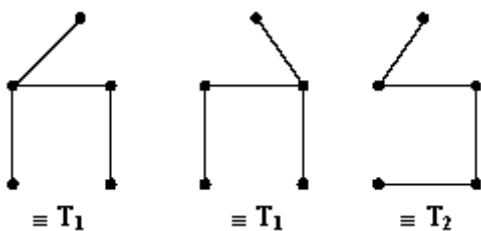
2

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Q2.

- (a)



M1

(for subgraphs)

A1× 3

6

- (b)

ft sensible answers

B1× 3

3

[9]

Q3.

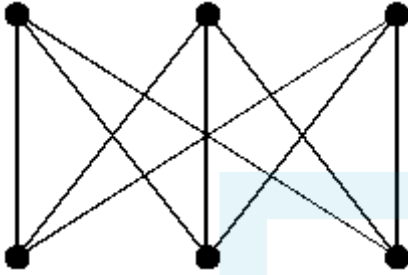
- (a) $A \leftrightarrow 1, B \leftrightarrow 5, C \leftrightarrow 6, D \leftrightarrow 4, E \leftrightarrow 3, F \leftrightarrow 2$
or equivalent

M1 mapping specified

A1 triangles to triangles

A1 all OK

- (b) e.g.



M1 graph specs OK

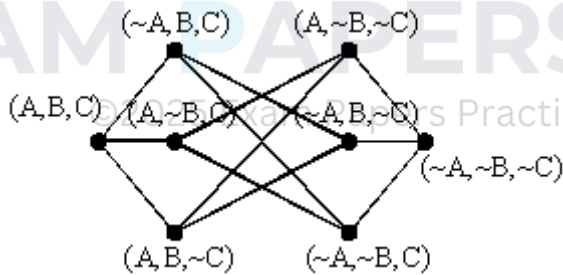
A1 not isomorphic to

5

[5]

Q4.

- (a)



(– 1 each error/omission)

M1

A2

3

Or with a 3 – D (cubical) graph.

- (b) e.g. (A,B,C) – 1
 (A,B,~C) – 2
 (A,~B,C) – 3
 (~A,B,C) – 4
 (~A,~B,C) – 5
 (~A,B,~C) – 6
 (A,~B,~C) – 7
 (~A,~B,~C) – 8

evidence of correct identification of vertices

M1

A1

2

(c) 6

M1 A1

(d) 6

M1 A1

4

[9]

Q5.

(a)



B1

(b)



B1

(c)



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M1

correct

A1

[4]

Examiner reports

Q2.

Candidates made very good attempts at constructing the three trees but, as one might expect, only the best were able to spot the isomorphic pairs.

Q4.

This question was answered well by candidates, but in part (b) some incorrectly thought that a degree count was sufficient to ensure isomorphism. A large proportion of candidates gave the correct answers to parts (c) and (d), but very few showed any working, thus giving themselves no opportunity to earn method marks should their counts have been wrong.

Q5.

Almost all candidates could correctly draw the graphs K_5 and $K_{3,3}$. Most reasoned that they were looking for a subgraph isomorphic to $K_{3,3}$ rather than K , in part (c), and were credited for that reasoning even if they could not find the required subgraph.